

Monte Carlo Simulation of Neutron Transmission Through Water Using OpenMC

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Abstract

This project investigated how water thickness affects neutron transmission using OpenMC 0.15.3 and a Python-based simulation workflow. A monoenergetic 1 MeV neutron pencil beam was directed perpendicular to a finite water slab. Five slab thicknesses, ranging from 1 to 20 cm, were evaluated using 300,000 neutron histories per case. Rear-face transmission decreased from approximately 87% at 1 cm to approximately 2% at 20 cm. A vacuum verification case reproduced the expected 100% transmission. A separate convergence study using up to three million histories showed stable transmission near 51% for the 5 cm slab and decreasing statistical uncertainty consistent with inverse-square-root scaling.

1. Objective

The objective was to develop a reproducible neutron-transport study that answered two questions:

1. How does increasing water thickness affect the fraction of incident neutrons exiting the slab's rear face?
2. How does the number of simulated neutron histories affect the statistical uncertainty of the transmission estimate?

The project combined material and geometry definition, Monte Carlo simulation, tally analysis, automated parameter sweeps, verification, and visualization.

2. Simulation Method

OpenMC follows simulated particle histories through a model, sampling interactions according to nuclear cross-section data. Tallies accumulate estimates of quantities such as particle current and reaction rates.

The model consisted of a water slab with a fixed 40 × 40 cm face and adjustable thickness along the x-axis. A source located in a vacuum gap ahead of the slab emitted neutrons in the positive x direction.

Parameter	Specification
Transport code	OpenMC 0.15.3
Calculation mode	Fixed source
Source	Normally incident neutron pencil beam
Source energy	1 MeV
Water composition	H1 and O16 in a 2:1 atomic ratio
Water density	0.997 g/cm ³
Water temperature	293.6 K
Slab face dimensions	40 × 40 cm
Slab thicknesses	1, 2, 5, 10, and 20 cm
Exterior boundaries	Vacuum
Nuclear data	Official OpenMC ENDF/B-VII.1 processed library
Thermal scattering	Hydrogen bound in water, c_H_in_H2O
Thickness-study histories	300,000 per case, in 30 batches

Vacuum boundaries removed particles once they exited the model. Consequently, neutrons could leave through the rear face, return through the entrance region, escape laterally, or be absorbed in the water.

Transmission Measurement

A surface-current tally measured neutron crossings through the rear vacuum boundary. Because particles could only cross this boundary outward, the tally represented rear-face transmission per source neutron:

$$T = \frac{\text{neutrons exiting the rear face}}{\text{source neutrons simulated}}.$$

The measurement included both uncollided neutrons and neutrons that scattered before exiting. It was not a measurement of radiation dose or exclusively uncollided attenuation.

Python automated the thickness changes, assigned random seeds, launched simulations, and extracted tally means and standard errors. Results were saved as CSV tables, and figures were generated using Matplotlib.

3. Verification

A vacuum case tested the source, geometry, and transmission tally. The water was temporarily removed while the geometry and directed source remained unchanged.

Every neutron was expected to travel directly to the rear face. The simulation produced:

Quantity	Result
Rear-face transmission	100.000000%
Reported standard error	0.000000 percentage points
Verification outcome	Pass

The zero sampling error was expected for this case because every particle history contributed the same transmission outcome.

This check supported the correctness of the basic source and tally configuration. It did not independently validate the nuclear interaction data or the water-model predictions against experiment.

4. Thickness-Study Results

Rear-face transmission decreased as water thickness increased.

Water thickness	Approximate transmission
1 cm	87%
2 cm	76%
5 cm	51%

10 cm	21%
20 cm	2%

Values in this table are approximate readings from the plotted results.

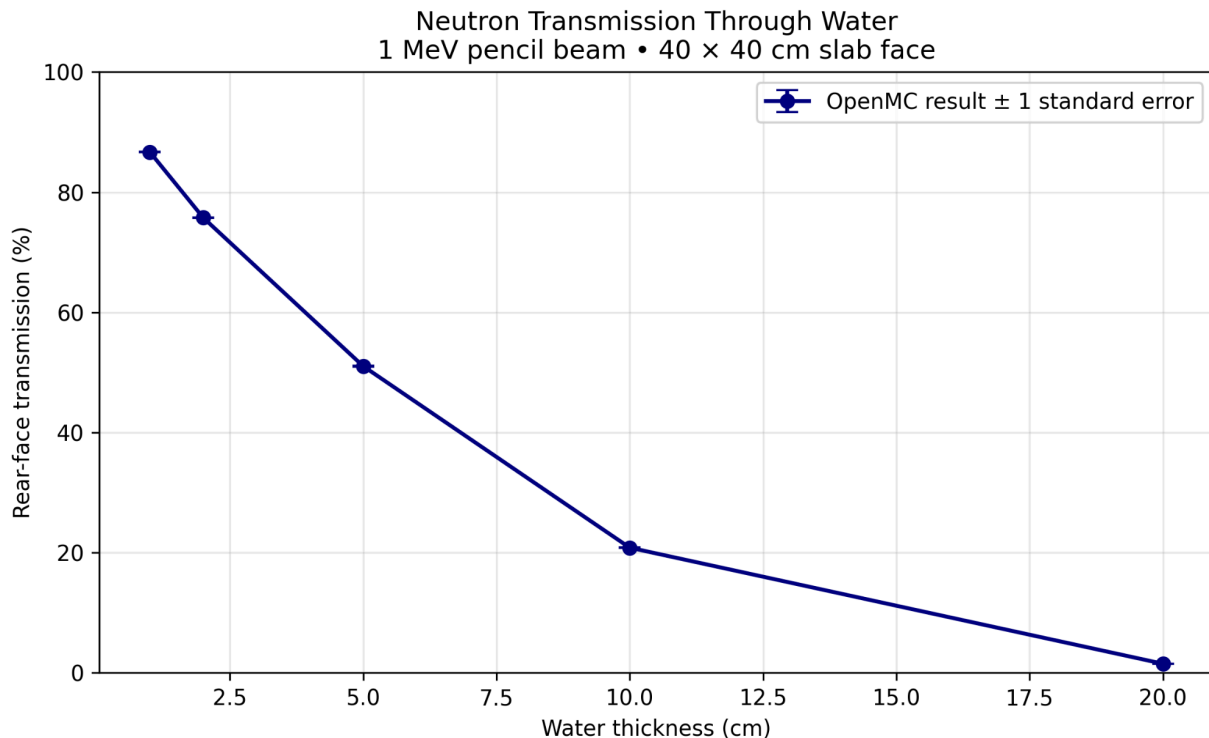


Figure 1. Rear-face neutron transmission versus water thickness for a 1 MeV pencil beam and a 40 × 40 cm slab face. Each calculation used 300,000 histories. Error bars show one standard error; connecting lines are visual guides.

The results show that increasing water thickness substantially reduced the fraction of neutrons reaching the rear face. However, the transmission tally did not identify the fate of the remaining neutrons. Reduced transmission could reflect absorption, escape through the front, or lateral leakage.

Because scattered neutrons were included in the tally, the results should not be interpreted directly using a simple exponential formula for uncollided transmission.

5. Statistical Convergence

A separate study held the water thickness at 5 cm and varied the total number of histories. Each calculation used 30 batches and a different random seed.

Neutron histories	Transmission	Standard error
30,000	50.960%	0.273 percentage points
300,000	51.027%	0.084 percentage points
3,000,000	51.020%	0.029 percentage points

The transmission estimates remained consistent as the number of histories increased. Increasing the history count by a factor of ten reduced the estimated standard error by factors of approximately 3.25 and 2.90, respectively.

These reductions were close to the expected Monte Carlo scaling:

$$\sigma_{\bar{T}} \propto \frac{1}{\sqrt{N}},$$

where N is the number of histories and $\sigma_{\bar{T}}$ is the standard error of the transmission estimate. A tenfold increase in histories therefore suggests an uncertainty reduction of approximately $\sqrt{10} \approx 3.16$.

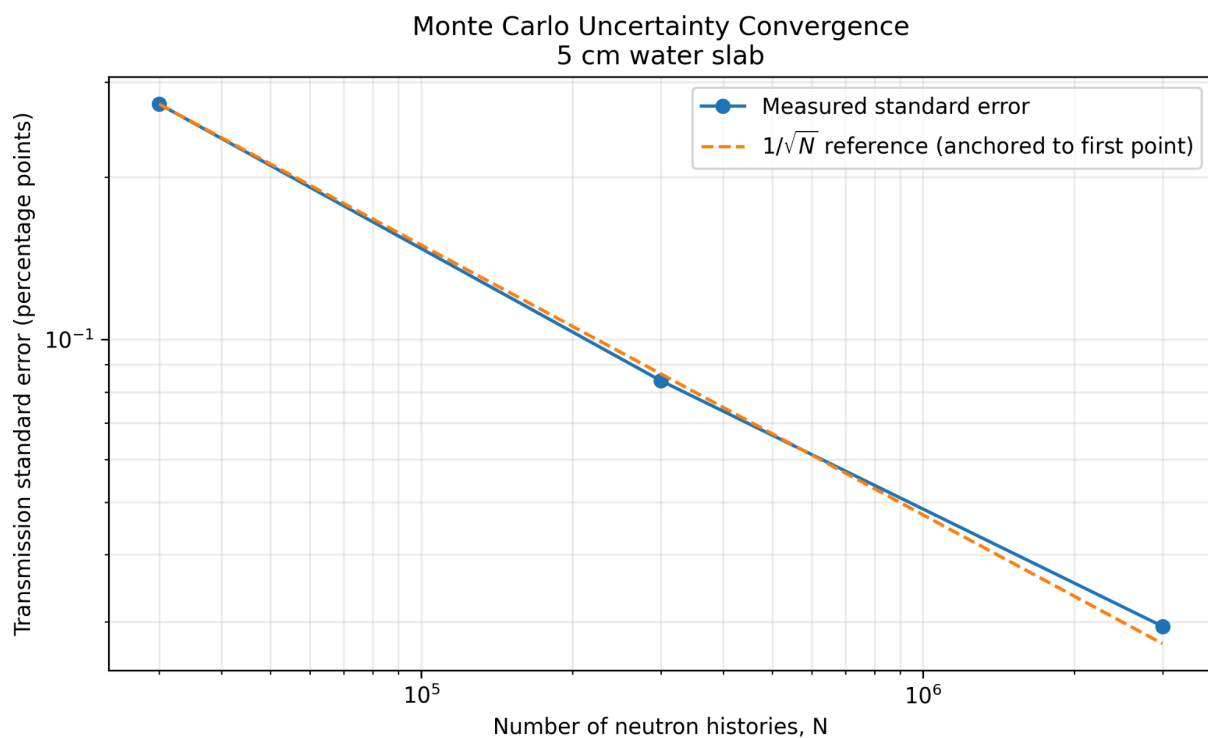


Figure 2. Transmission standard error versus neutron-history count for the 5 cm water slab. The reference curve follows $\frac{1}{\sqrt{N}}$ and is anchored to the first measured error.

At three million histories, the estimated transmission was **51.020% ± 0.029 percentage points**, with uncertainty reported as one standard error. The observed convergence supported the numerical stability of the estimate under the selected model assumptions.

6. Reproducibility and Data Preparation

The workflow was organized into a nuclear-data preparation notebook and a main simulation notebook.

During initial setup, the existing nuclear-data index was found to be empty, and some data files were unreadable. A fresh official ENDF/B-VII.1 archive was downloaded. The H1, O16, and hydrogen-in-water thermal-scattering datasets were extracted, loaded successfully using OpenMC's data readers, and registered in a new index.

The main notebook contained the model definition, baseline calculation, vacuum verification, thickness sweep, convergence study, and figures. Simulation cases were saved in separate directories, preserving input files and statepoint results.

After removing obsolete troubleshooting cells, the main notebook was successfully executed from a restarted kernel without errors. This demonstrated that the analysis did not depend on variables retained from earlier interactive work.

7. Limitations

The study used an idealized monoenergetic pencil beam rather than a distributed source with a continuous energy spectrum. The finite slab dimensions allowed lateral leakage, so the results depend on the specified geometry.

Reported standard errors describe Monte Carlo sampling uncertainty only. Uncertainty in nuclear data, material properties, and modeling assumptions was not quantified.

Absorption and leakage through other faces were not tallied separately. The study therefore did not establish a complete neutron balance. No comparison with experimental measurements or an independent transport code was performed.

The results characterize neutron transmission for this model and do not establish radiation-dose reduction or shielding performance for a specific facility.

8. Conclusions and Future Work

The project demonstrated a complete Python-driven OpenMC workflow for studying neutron transmission through water. Rear-face transmission decreased from approximately 87% to 2% as slab thickness increased from 1 to 20 cm.

The vacuum case reproduced the expected 100% transmission, and the convergence study showed a stable 5 cm transmission estimate near 51% with statistical uncertainty decreasing approximately as $\frac{1}{\sqrt{N}}$.

The next extension is a neutron-balance study that separately measures absorption and leakage through each exterior face. Additional work could examine transmitted energy spectra and compare predictions with a suitable published benchmark.

References

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